JAMES RUSE AGRICULTURAL HIGH SCHOOL

2024

YEAR 10 SCIENCE TEST

YEARLY PAPER

MARKING CRITERIA

Total: 80 Marks

Reading time – 5 minutes

Working time – 90 minutes

Answer questions in the spaces provided.

General Instructions

- NESA Board-approved calculators may be used.
- Write using a black pen.
- Draw diagrams using a pencil.
- Data Sheet and periodic table provided.

1. Which evidence best supports the theory that the universe was created by an explosion called the Big Bang?

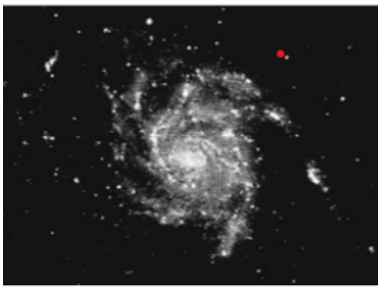
   A. Impact craters found on Earth.

   B. Cosmic background radiation.

   C. The different compositions of terrestrial and gaseous planets.

   D. The blue shift of light from distant galaxies.

2. The photograph below shows the universe as seen through a telescope.



This feature is best identified as

   A. A galaxy

   B. A comet

   C. An asteroid

   D. A star

3. Which process generates the energy that is ultimately released by stars?

A. Nuclear fusion

B. Convection currents

C. Thermal conduction

D. Radioactive decay

4 What celestial phenomenon most affects radio communication and other electrical atmospheric changes for us on Earth?

A. Solar eclipse

B. Solar flares

C. Lunar eclipses

D. Meteorites entering the atmosphere

5. What causes clouds of dust and gas to form a protostar?

A. Magnetism

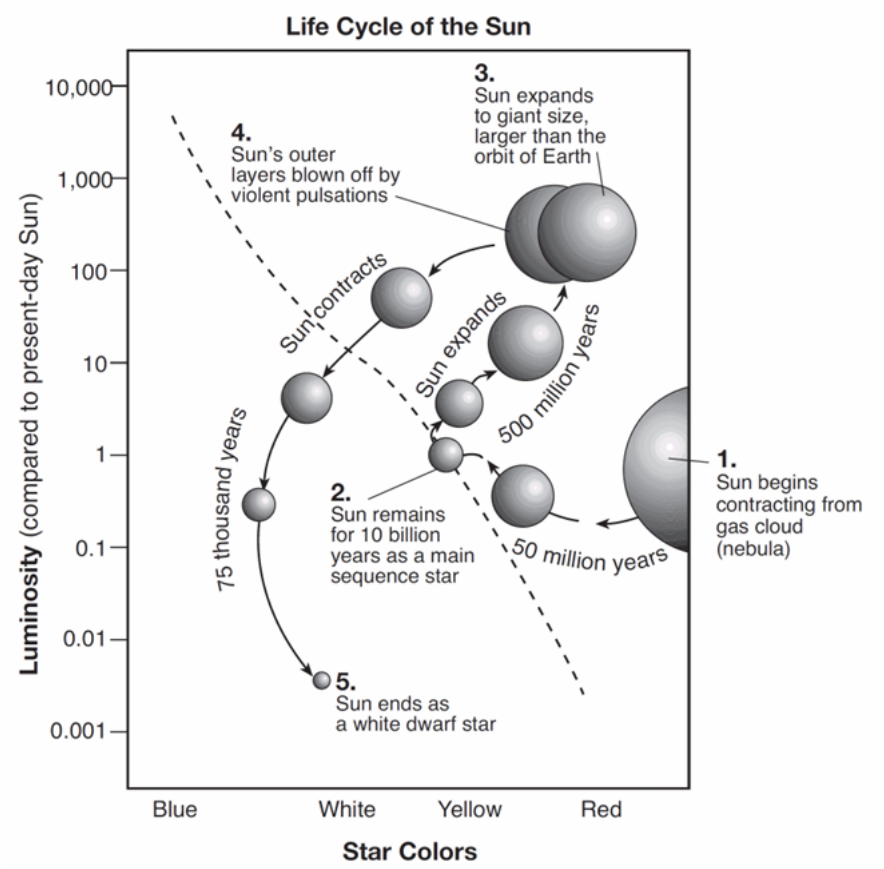
B. Gravitational attraction

C. Expansion of matter

D. Cosmic background radiation

6. Base your answer to the following question on the diagram below.

The diagram represents the inferred changes to the luminosity and colour of the Sun throughout its life cycle. The diagonal dashed line represents the main sequence stars. The numbers 1 through 5 represent stages in the life cycle of the Sun.



The Sun is inferred to be the most luminous when it is classified as a

A. White dwarf star

B. Gas cloud (nebula)

C. Main sequence star

D. Giant star

7. A student is reacting hydrochloric acid with zinc granules in a beaker. Which of the following would increase the rate of this reaction?

A. Decreasing the temperature.

B. Adding water to the beaker.

C. Using a test tube instead of a beaker.

D. Cutting the zinc into smaller pieces.

8. Which of the following best defines corrosion?

A. The process by which a metal reacts to form a metal oxide.

B. The gradual breakdown of a metal due to chemical reactions with the environment.

C. The process of metals is becoming more conductive over time due to exposure to air.

D. The formation of rust on iron after exposure to salt and water.

9. Two islands are drifting apart due to geological changes. When the islands were close together, similar populations of an insect were found on each island.

As the islands drift further apart, which process must happen for speciation to occur?

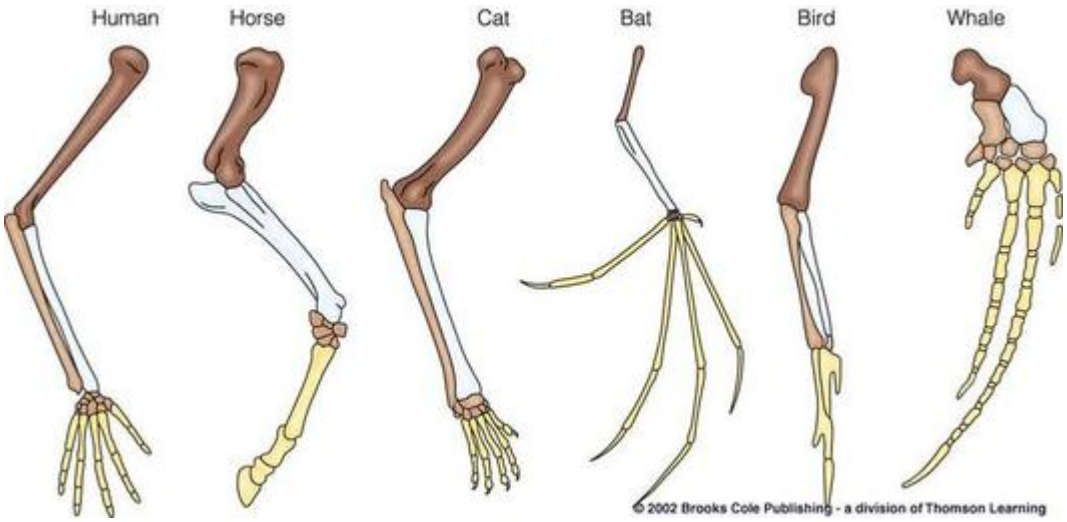
A. Different predators evolve.

B. Reproductive isolation.

C. Climate change.

D. Continental drift.

10. Consider the following skeletal structures:



Which type of evolution does this illustrate?

A. Codominant

B. Dominant

C. Convergent

D. Divergent

11. For natural selection to take place, many more offspring must be born than survive to maturity.

What else is required?

A. Inheritable characteristics that suit the environment.

B. A decrease in the number of predators.

C. Characteristics of the ecosystem change.

D. A mass extinction event occurs.

12. There is much more phenotypic variation among domesticated dog breeds than in the closely-related non-domesticated coyote.



What accounts for this observation?

A. Higher rates of mutation during domestication.

B. Random mating during domestication.

C. Domesticated dogs would have a low survival rate in the wild.

D. Domesticated dogs exhibit a high rate of growth.

13. Which of the following lists layers of the atmosphere correctly for decreasing altitude?

A. Troposphere, stratosphere, exosphere

B. Thermosphere, stratosphere, mesosphere

C. Exosphere, troposphere, thermosphere

D. Thermosphere, mesosphere, stratosphere.

14. La Niña is associated with which of the following conditions?

A. Warmer east Pacific temperatures and low trade winds.

B. Warmer east Pacific temperatures and high trade winds.

C. Cooler east Pacific temperatures and low trade winds

D. Cooler east Pacific temperatures and high trade winds

15. Which of the following can result from reduced ozone levels in the stratosphere?

A. Increase rate of droughts.

B. Increased rate of genetic mutation

C. Rising sea levels

D. More frequent tsunamis

16. Which of the following processes directly decreases the amount of phosphorus in the biosphere?

A. Mining

B. Erosion

C. Weathering

D. Excretion

17. What is a fertilised egg cell known as?

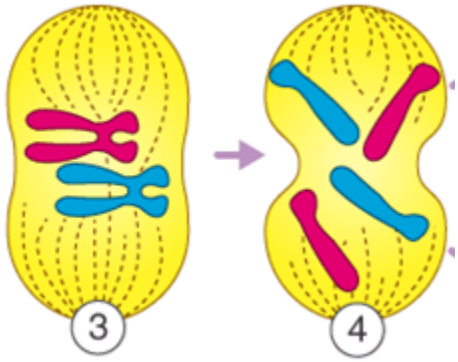
A. Gamete

B. Germ cell

C. Zygote

D. Ovum

18. The following image shows 2 stages of mitosis.



What is the purpose of the spindle fibres?

A. To provide support for the chromatids.

B. To duplicate the chromatid.

C. To pull the chromatids apart.

D. To attach the sister chromatids together.

19. Which species is being reduced in the following reaction?

$4\ Al + 3\ Fe_2O_3 \rightarrow 2\ Al_2O_3 + 6\ Fe$

A. Aluminium

B. Iron (III) oxide

C. Aluminium oxide

D. Iron

20. Which of the following is an example of an endothermic reaction?

A. Photosynthesis

B. Rusting of iron

C. Combustion of fuel

D. Respiration

Name:

Class: **10A 10C 10K 10L 10T**

ANSWER BOOKLET

Section A Answer grid for multiple choice questions 1 – 20.

1.	A O	B O	C O	D O
2.	A O	B O	C O	D O
3.	A O	B O	C O	D O
4.	A O	B O	C O	D O
5.	A O	B O	C O	D O
6.	A O	B O	C O	D O
7.	A O	B O	C O	D O
8.	A O	B O	C O	D O
9.	A O	B O	C O	D O
10	A O	B O	C O	D O
11.	A O	B O	C O	D O
12.	A O	B O	C O	D O
13.	A O	B O	C O	D O
14.	A O	B O	C O	D O
15.	A O	B O	C O	D O
16.	A O	B O	C O	D O
17.	A O	B O	C O	D O
18.	A O	B O	C O	D O
19.	A O	B O	C O	D O
20.	A O	B O	C O	D O

SECTION B (60 marks)

Answer the questions in the spaces provided.

These spaces guide the expected length of response.

Show all relevant work in questions involving calculations.

Extra writing space is provided at the back of this booklet.

If you use this space, clearly indicate which question you are answering.

Question 21 (3 marks)

Describe the role of microorganisms in the nitrogen cycle.

Marking Criteria	*Mark(s)*
● Describes the role of microorganisms in the nitrogen cycle	3
● Outlines some roles of microorganisms in the nitrogen cycle	2
● Identifies a function of microorganisms in the nitrogen cycle	1

Bacteria play an important role in converting inorganic nitrogen such as atmospheric N_2 into forms available to be taken up by plants such as nitrate, nitrite and ammonium ions through the process of nitrification. Some bacteria are free-living in the soil, while others are found in nodules in plants such as legumes.

Bacteria and fungi act as decomposers to break down dead material into simpler substances. Denitrifying bacteria can convert nitrogenous compounds back into the elemental form, removing it from the biosphere and returning it to the atmosphere.

Marking comments: many answers were very vague and did not give details about the different roles. At least three different roles were needed for full marks (nitrogen fixing, nitrifying, denitrifying and biological decomposition)

Question 22 (2 marks)

Outline how temperature and salinity drive the formation of deep ocean currents. **2**

Marking Criteria	*Mark(s)*

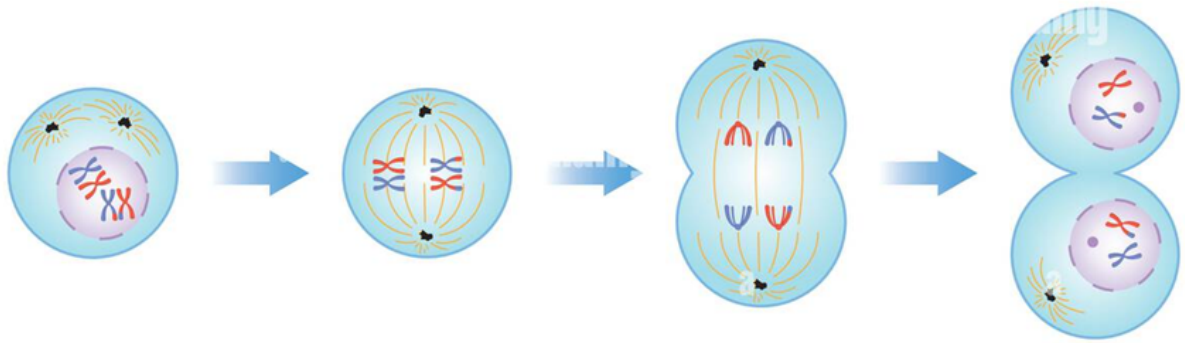
• Relates temperature and salinity to density • Outlines a cause for high salinity at the poles • Relates increased density to deep ocean currents	2
• Outlines one factor that affects deep ocean currents	1

Deep ocean currents are formed by dense water sinking, with other water being dragged in to replenish it. Near the South pole, water starts to freeze, leaving behind water with higher salinity, increasing density. Near the North pole, winds cause high evaporation resulting in high salinity, increasing density. The low temperature of the water also means that it is more dense than warmer water. Hence, near the poles, sinking water results in the formation of deep ocean currents.

Marking comments: many students were able to relate temperature and salinity to density, but very few were able to give a good reason why the water would be both cold and salty at the same time.

Question 23 (5 marks)

The following diagram shows some of the stages of cell division.



(a) Identify the stage in which the chromosomes line up in the centre of the cell. **1**

Metaphase

(b) Justify why this must be meiosis and not mitosis. **1**

- Crossing over of genetic material has occurred in prophase (as seen with the mix of red and blue on the chromatids). OR
- Homologous chromosomes line up in pairs in metaphase instead of in a line like in mitosis OR
- In anaphase the homologous pairs are pulled to opposite ends of the cell, whereas in mitosis the

chromatids are pulled apart.

Note: Some students gave reasons why it was meiosis which could also apply to mitosis. You needed to prove that it could only be meiosis. Some also referred to stages of meiosis II which are not shown in the diagram. Also, the cells in telophase do not have half as many chromosomes as the original – the original cell is not shown, but duplicates its chromosomes during interphase (before prophase).

Marking Criteria	*Mark(s)*
• Provides a reason why it must be meiosis and not mitosis	1

(c) Complete the following table. **3**

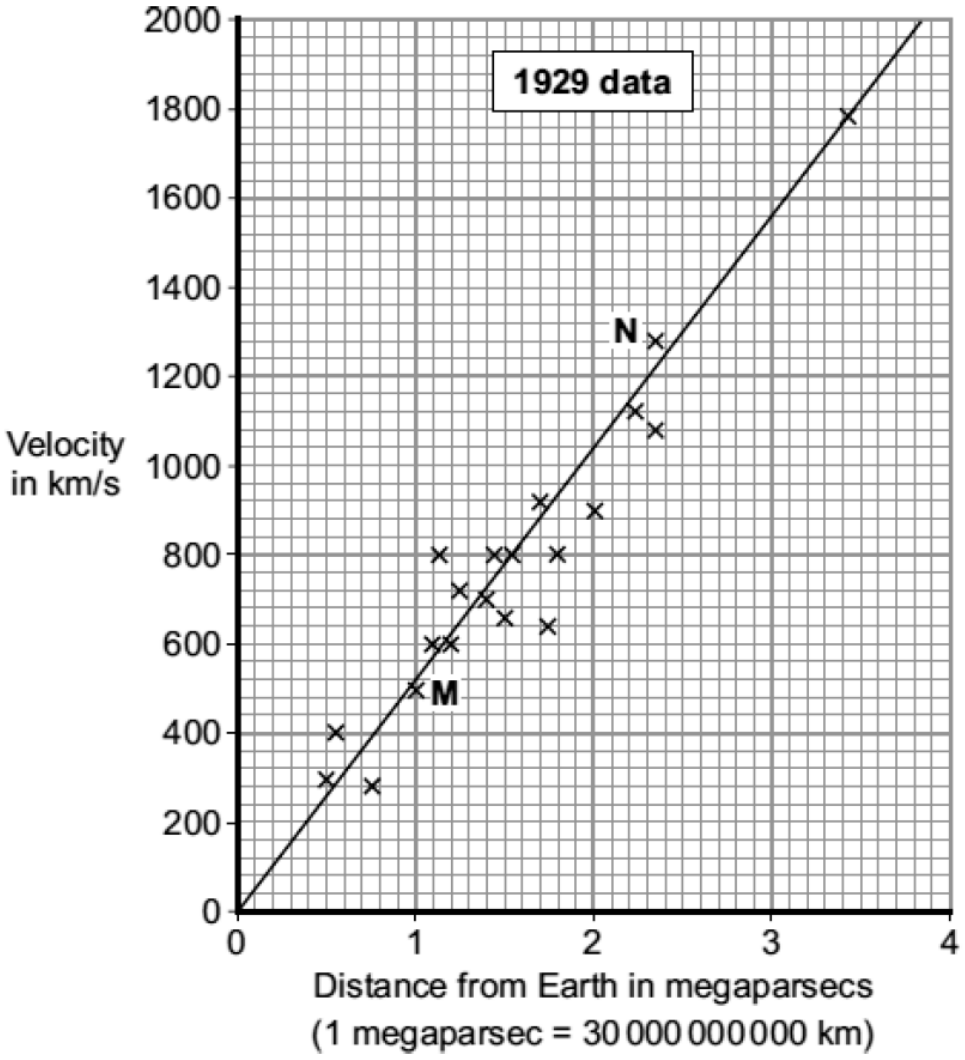
	Mitosis	**Meiosis**
Type of cells produced	somatic (accepted body cells, non-sex cells, genetically identical daughter cells)	Gametes or sex cells
Diploid/haploid	diploid	haploid
Purpose in humans	growth and repair	reproduction

Marking Criteria	***Mark(s)***
• 3 rows correct	3
• 2 rows correct OR 3 rows correct but swapped left/right columns	2
• 1 row correct	1

Question 24 (6 marks)

By measuring the *red-shift*, Hubble was able to calculate the speed at which the galaxies are moving away from the Earth. He was also able to calculate the distance of these galaxies from the Earth.

The graph shows the relationship between the data collected by Hubble.



Question 24 continues on Page 15

(a) The gradient of the line drawn on the graph gives a number known as the Hubble constant. The Hubble constant can be used to estimate when the universe began.

Use the graph to calculate the value of the Hubble constant.

Show clearly how you obtained your answer.

Hubble constant = _____km /s per megaparsec **2**

Marking Criteria	*Mark(s)*
● 520 with working (510 and 530 inclusive)	2
● Any relevant information	1

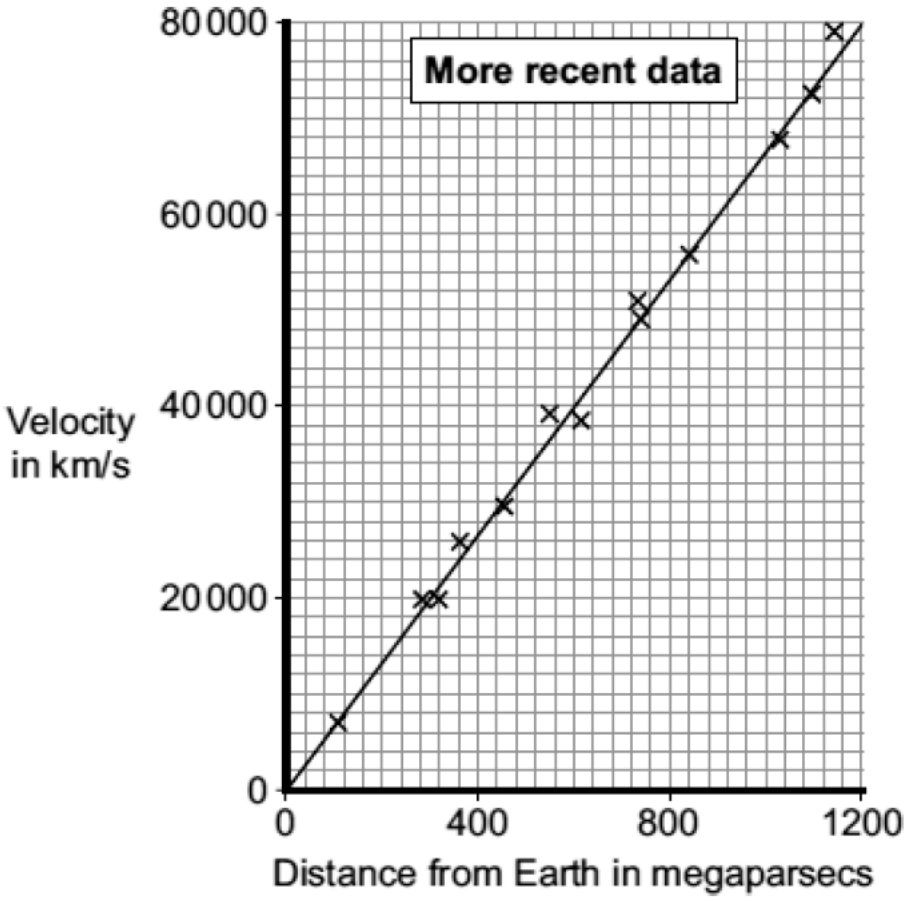
(b) The Andromeda galaxy is not moving away from the Earth. It is actually moving towards the Earth. This means that the light from Andromeda shows a blue-shift.

How do the wavelength and frequency of the light from Andromeda seem to have changed when viewed from the Earth? **2**

Marking Criteria	*Mark(s)*
● wavelength is decreased ● frequency is increased	2
● Any relevant information	1

Question 24 continues Page 16

(c) More recently, data has been obtained from more distant galaxies.



The results from the more recent data give a totally different value for the Hubble constant to the one calculated from the 1929 data.

Which set of data, the 1929 or the more recent, is most likely to give the value closest to the true value for the Hubble constant?

Draw a ring around your answer.

1929 **more recent**

Give a reason for your answer. **2**

Marking Criteria	*Mark(s)*

● more recent ● *no mark for this but must be given to gain reason mark* ● data more reliable ● *accept data is more accurate* ● **or improved equipment / techniques** ● *more technology is insufficient* ● **or data obtained from more (distant) galaxies** ● *accept a wider range of data* ● *accept data closer to the line of best fit* ● ***or data less scattered*** ● *accept no anomalous result*	2
● Any relevant information	1

Question 25 (4 marks)

The ‘steady state’ theory was once a popular alternative to the ‘big bang’ theory.

The ‘steady state’ theory suggested that the universe, although expanding, had no origin and it has always existed. As the universe expands, a small amount of matter is created to keep the universe looking exactly the same all of the time.

(a) When considering the origin of the universe, what is the difference between the ‘big bang’ theory and the ‘steady state’ theory? **3**

Marking Criteria	*Mark(s)*
● big bang theory – universe started at one point *(*then expanded*)* ● steady state theory – universe has no origin / has always existed ● *accept an answer in terms of mass* ● *eg steady state theory mass is created*	2
● Any relevant information	1

(b) The ‘steady state’ theory was important in encouraging new research into the universe.

Suggest a reason why scientists were keen to carry out new research. **1**

Marking Criteria	*Mark(s)*
● to find evidence to support one or both theories ● *accept prove for support* ● *accept to gain more knowledge about the universe* ● **or** to find evidence to disprove one or both theories	2
● Any relevant information	1

Question 26 (3 marks)

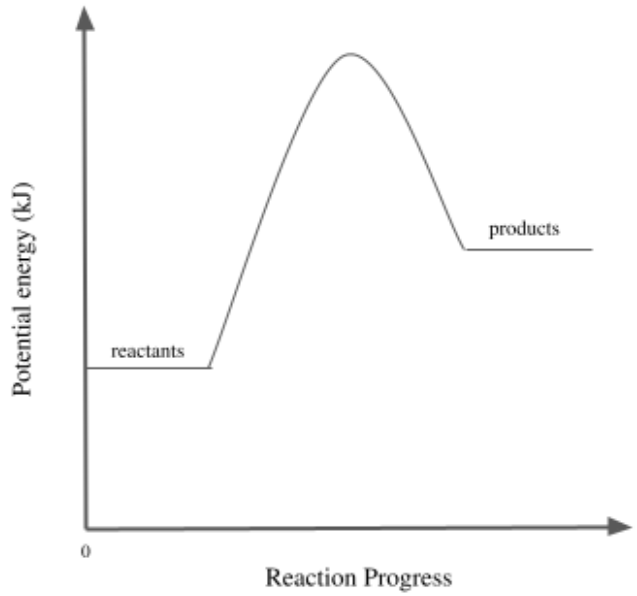
Explain the effect of decreasing concentration on a chemical reaction using collision theory. **3**

Sample answer:
According to collision theory, successful collisions are required for a reaction to occur. Decreasing the concentration of a reactant decreases the number of particles per volume which means there are less particles to collide, which decreases the rate of reaction.

Marking Criteria	*Mark(s)*
• Explains that decreasing concentration decreases particle collisions • Identifies that reaction rate decreases • Refers to collision theory stating that successful collisions are required for a reaction	3
• 2 of the above	2
• Provides some relevant information	1

Question 27 (3 marks)

The following diagram shows the energy profile for a reaction.

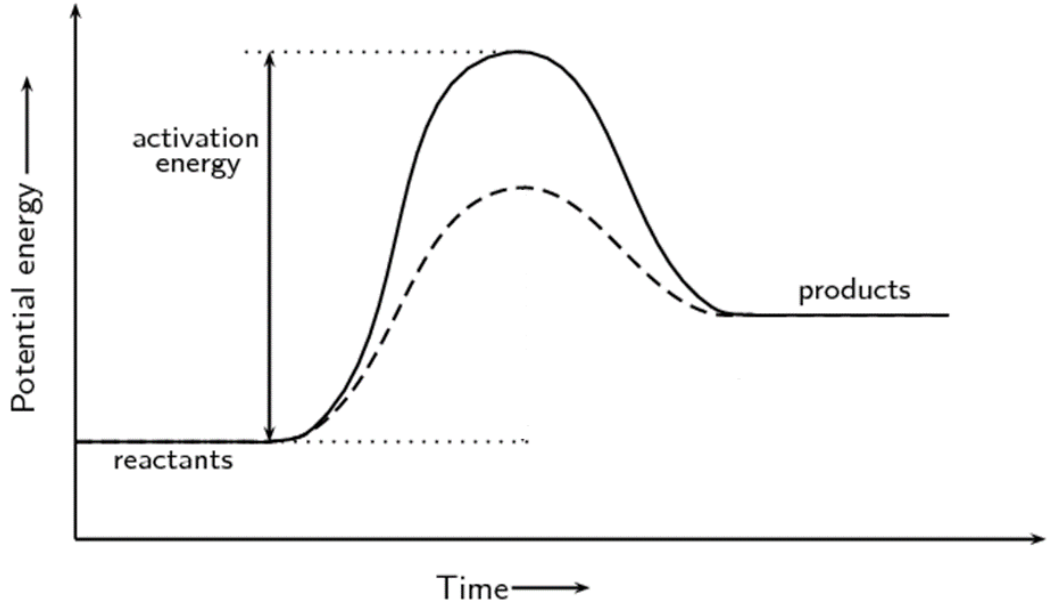


(a) Identify what type of reaction this would be in terms of energy transfer? **1**

endothermic

(b) Label on the diagram:

- The activation energy for the reaction **1**
- A possible energy profile if a catalyst was added. **1**



Question 28 (3 marks)

Outline the role of index fossils in understanding sequences of events in evolution. **3**

Marking Criteria	*Mark(s)*
• Defines properties of an index fossil • Shows how index fossils can be used to date other fossils • Relates this to evolutionary sequence	3
• Two of the above	2
• Some relevant information	1

Index fossils are remains or traces of species that lived over a wide geographic distribution during a well-defined limited time range. The presence of an index fossil in a rock sample can date other fossils in the same rock layer. Because sedimentary rock forms in layers with the youngest rocks on

top and the oldest at the bottom, this can give information about the evolutionary order of species, and which existed together at the same time.

Question 29 (3 marks)

Outline how a volcanic eruption can have an effect on the Earth's spheres. **3**

Marking Criteria	*Mark(s)*
● Outlines effect of volcanic eruption on at least two spheres	3
● Outlines effect of volcanic eruption on one sphere ● OR ● Identifies effects of volcanic eruption on at least two spheres	2
● Some relevant information	1

The 1991 volcanic eruption at Mount Pinatubo in the Philippines resulted in large clouds of ash released into the stratosphere. This ash cloud travelled around the world several times. The amount of light reaching the Earth's surface reduced by up to 10 % for a year, causing surface temperatures to be between 0.2 and 1.0 °C lower than average for the next year. This reduced the amount of photosynthesis, affecting the biosphere of the Earth. (could also include the formation of acid rain or the destruction of the ozone layer).

Question 30 (2 marks)

Give the chemical names of TWO gases whose emissions makes major contributions to the enhanced greenhouse effect. **2**

Marking Criteria	*Mark(s)*
● Gives the names of two species	2
● Gives the name or formula of one species	1

Two of the following: Carbon dioxide, methane, nitrous oxide

Question 31 (6 marks)

A student set up 2 beakers as follows:

Beaker 1 – Nickel metal in magnesium sulfate solution.

Beaker 2 – Magnesium metal in nickel sulfate solution.

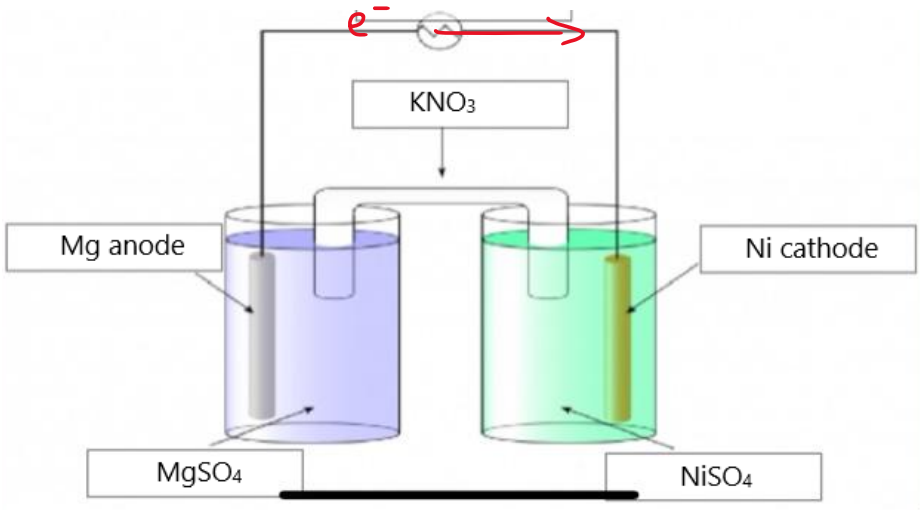
They observed a metal being deposited in one of the beakers and no reaction in the other.

(a) Using the reactivity series of metals, identify which beaker had the reaction occurring. **1**

Beaker 2

Explanation: Looking at the back of your periodic table, magnesium is higher than nickel and so more reactive (i.e. the reverse reaction occurs – Mg □ $Mg^{2+} + 2e^-$ This means the magnesium is dissociating into ions in the solution, and nickel ions are turning into nickel metal ($Ni^{2+} + 2e^-$ □ Ni).

The student then set up a galvanic cell according to the following diagram.



(b) Draw an arrow on the diagram showing the direction of electron flow. **1**

Arrow from the anode to the cathode via the wire, not the salt bridge.

(c) Outline the purpose of the KNO_3 salt bridge. **1**

Facilitates the flow of ions between the 2 half-cells, maintaining electrical neutrality.

Note: K^+ and NO_3^- ions (not electrons) flow from the salt bridge to prevent the build-up of negative charge from the electrons.

Marking Criteria	*Mark(s)*

• Outlines the purpose of the salt bridge (i.e. electrical neutrality, or at least mentions the transfer of ions)	1

(d) Write a balanced chemical equation for the reaction of this cell, including states. **3**

$Mg\,(s) + NiSO_4\,(aq) \rightarrow MgSO_4\,(aq) + Ni\,(s)$

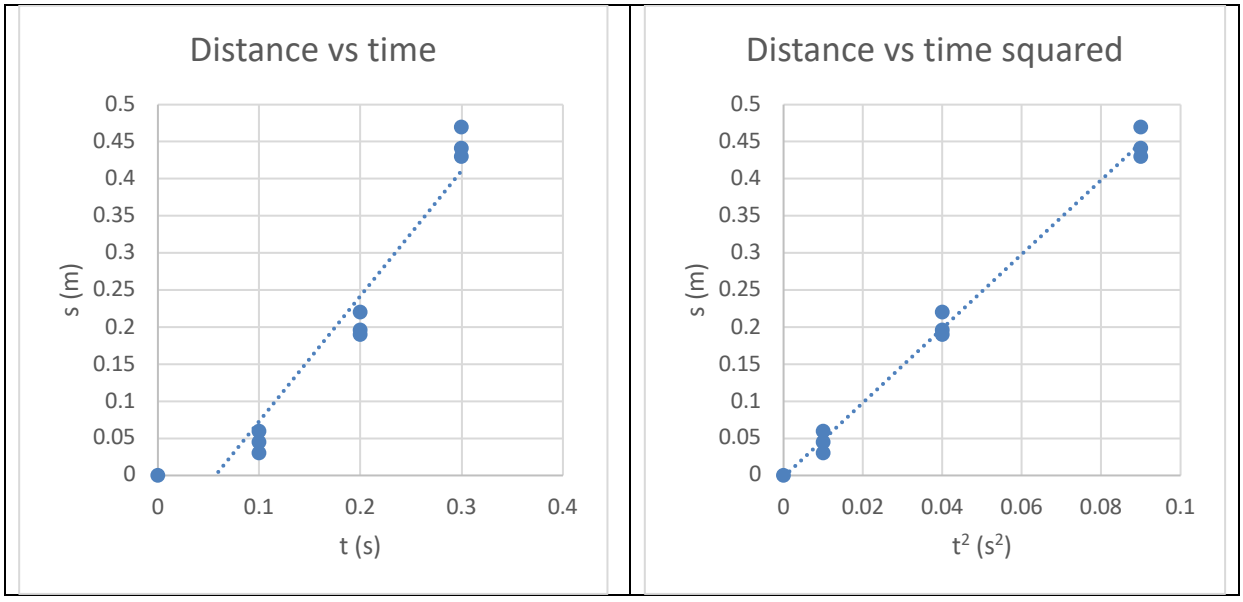
Note: A few students swapped the reactants and products – if I drew a backwards arrow, that doesn't mean you can write reactions backwards, it was just to indicate you needed to swap them.

Marking Criteria	*Mark(s)*
• Correct balanced chemical equation with states	3
• Makes 1 error (e.g. wrong states, wrong species, unbalanced, swapped reactants and products)	2
• Makes 2 errors	1

Question 32 (10 marks)

In order to determine a relationship between variables data needs to be processed so that a straight line can be drawn through the points.

For example, to show that the distance travelled by a falling ball is proportional to time squared, the values for time would be squared, and a graph of s vs t^2 would be checked for a straight line:

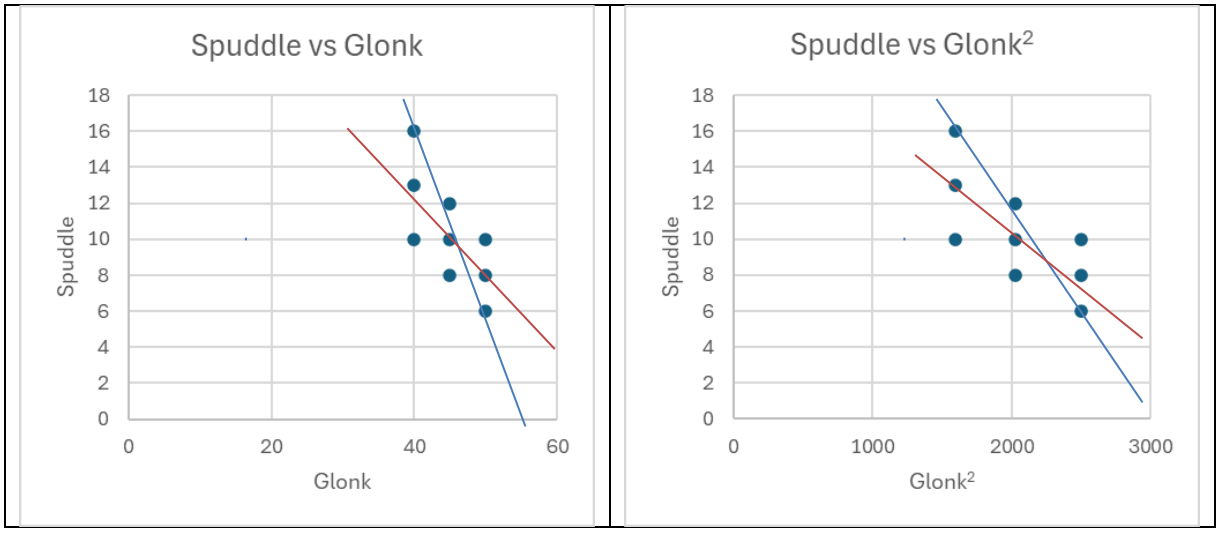


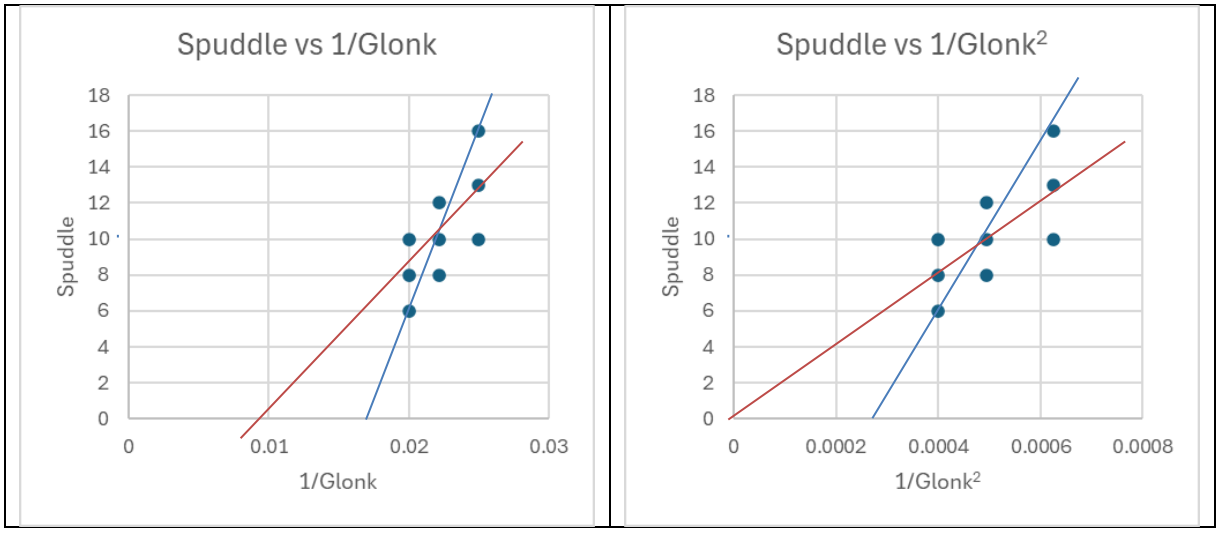
Here a straight line through the origin can not be drawn through the data. This is inconsistent with the relationship s proprortional to t. As a horizontal line cannot be drawn through the data, it is also inconsistent with no correlation between s and t.	Here a straight line through the origin can be drawn through the data. This is consistent with the relationship s is proportional to t^2.

The graph also shows that each data point has been repeated for a total of three times, giving a spread of data. When there is a spread of data, the line is considered to go through the data point if it passes through the range of the data points if there is less than 5 repeats. If there are 5 or more repeats, the line should go somewhere through the average ± 1 standard deviation. Given the spread of data shown above, a horizontal line cannot be drawn through the data points for the unprocessed data (s vs t graph), so there is some correlation between s and t.

Question 32 continues on page 22

Caspar from the planet Zorg is investigating the relationship between the variables glonk (G) and spuddle (S). Data is collected with three repeats, and a range of graphs plotted.





Question 32 continues on page 23

(a) What conclusion(s) could Caspar draw about the relationship between spuddle and glonk?

Annotate the graphs of spuddle and glonk to support your response. **4**

Marking Criteria	*Mark(s)*
• Annotates each graph at least once • Shows that the data is consistent with no relationship, inverse law and inverse square law • Shows that the data is inconsistent with a linear relationship and squared relationship	4
• As above with 1-3 errors or omissions	3
• As above with 4-6 errors or omissions	2

• Some relevant information	1

The line of best fit is shown in red for each, and possible straight lines for each are bounded by the two blue lines.

For spuddle vs glonk, the line can not go through the origin. The data is inconsistent with spuddle is proportional to glonk.

For spuddle vs 1/glonk, a horizontal line can be drawn. The data is consistent with no correlation between spuddle and glonk.

For spuddle vs glonk^2, the line can not go through the origin. The data is inconsistent with spuddle is proportional to glonk^2.

For spuddle vs 1/glonk, the line could go through the origin. The data is consistent with spuddle is proportional to 1/glonk.

For spuddle vs 1/glonk^2, the line could go through the origin. The data is consistent with spuddle is proportional to 1/glonk^2. The line of best fit goes through the origin, this is the best match to data for all the relationships tested.

(b) As an earthling, explain to Caspar how the data collection could be improved to better understand the relationship between spuddle and glonk.

Give THREE different suggestions. Higher marks will be awarded to better suggestions. **6**

Suggestion 1

……

……

…….

Suggestion 2

……

……

…….

Suggestion 3

……

……

…….

Marking Criteria	*Mark(s)*
● Makes 3 significant suggestions ● Uses cause and effect statement to show how the change will strengthen the conclusion	5-6
● Makes 2 significant suggestions ● Outlines how the changes will strengthen the conclusion ● OR ● Makes 3 suggestions which will improve the data analysis ● Outlines how the changes will strengthen the conclusion	3-4
● Makes 2 significant suggestions ● OR ● Outlines how one significant suggestion would strengthen conclusion	2

● OR ● Makes 3 suggestions which will improve the data analysis	
● Some relevant information	1

Answers could include:

Increase the range of the data points to smaller values of glonk. There isn't much of a range to distinguish between 1/glonk and 1/glonk^2 as the points are quite close together. It might also remove the no correlation conclusion if the lowest data point for a smaller value was higher than 10.

Increase the number of repeats for each data point. This will help to reduce random error, bringing the acceptable range through which the line(s) could go, potentially removing the no correlation between spuddle and glonk as a horizontal line is just possible with three points.

Increase the sensitivity of the measuring equipment. All glonk values are integers, if the measuring instrument were to have finer divisions, it might be easier to rule out the no correlation line.

Marking comments: there was no indication as to the order of data collection. Many students did not give suggestions relating to data collection.

Question 33 (5 marks)

Discuss how TWO different types of evidence (other than index fossils) can be used to support our understanding of evolution. Use at least one example of each to support your response. **5**

Marking Criteria	*Mark(s)*
● Provides points for and/or against two types of evidence ● Relates these types of evidence to evolutionary theory ● Supports discussion with at least one example for each type of evidence	5
● As above, but missing 1-2 points	4
● Outlines two pieces of evidence	3

● Relates these to evolutionary theory ● Provides one example of evidence	
● Identifies two types of evidence ● OR ● Identifies one type of evidence AND EITHER an example OR relates it to evolutionary theory	2
● Some relevant information	1

Answers could include:

Biochemical evidence uses the similarities of molecules such as proteins to identify how similar two species are. As the number of changes in a protein sequence increases over time, organisms with similar sequences for a highly conserved protein such as cytochrome c are believed to be more closely related than those pairs with less similar sequences. Biochemical evidence can reveal relationships that might be masked by significant differences in physical appearance. However, as protein molecules don't generally persist over time, it is difficult to use this for extinct organisms, so evolutionary pathways need to be inferred.

Comparative anatomy involves comparing structural features of organisms to estimate how recently they evolved from a common ancestor. A well-known example is the pentadactyl limb of vertebrates, which has taken on many forms such as a human hand, a bird's wing and a whale's flipper. This suggests that a common ancestor had a pentadactyl limb, but due to different ecological niches, the structure evolved to result in different functions, in a process termed divergent evolution. However, it is important to note that convergent evolution, where similar structures evolve to meet the same function such as wings for birds and butterflies, are distinguished, as their last common ancestor was a considerably longer time ago and did not necessarily have the common structure of wings.

Biogeography examines the spatial distribution of extinct species as well as related extant species. One example of this is marsupials, found in both Australia and the Americas, continents which have a substantial ocean between them. Fossil records of marsupials are also found in Antarctica, consistent with the theory of plate tectonics in which Australia, South America and Antarctica (along with other landmasses such as Africa) were once joined as part of Gondwana. Incomplete data, however, means that extrapolation must be used to infer evolutionary relationships.

Question 34 (5 marks)

"Biotechnology holds transformative potential, but it is crucial that we address ethical concerns proactively. The impact of our innovations on society, the environment, and individual rights must be considered as we push the boundaries of what is possible."
– George Church

Evaluate the use of one of the following technologies and its impact on society. **5**

- Cloning
- DNA profiling
- Gene therapy
- Genetic screening
- Stem cells
- Genetic modification

DNA profiling is a process used to identify an individual's unique genetic makeup. A biological sample such as hair, saliva or blood is taken from a person and sections of DNA are analysed to create a unique profile for that individual. It has various applications in forensics and medicine.

Benefits of DNA profiling include using it to match DNA samples from a crime scene to a suspect to identify criminals, paternity testing to determine the father of an individual, and in medical research to check for predisposition for particular diseases.

Limitations of the technology include requiring enough material for an accurate sample – some crime scenes may not have enough to provide accurate analysis. There may also be errors in the processes involved in developing the profile such as extracting and analysing the DNA. There are also ethical concerns regarding privacy of an individual's medical/genetic information, as well as potential for discrimination based on a person's genetic makeup.

Overall, the use of DNA profiling has been of great benefit to society by helping to solve crime as well as providing advancements in medicine. The privacy concerns can be easily mitigated by maintaining confidentiality of the data.

Notes – many students listed advantages and disadvantages but didn't provide enough detail of how the technology actually works, i.e.:

- cloning – an egg cell has its nucleus removed and replaced with another nucleus from the organism being cloned, which then grows into that
- DNA profiling – DNA is taken from biological samples such as hair, saliva or blood and then sections are analysed
- Gene therapy – sections of genes are altered or replaced using technology such as viral vectors, injections, or CRISPR-Cas9

- Genetic screening – DNA is analysed to screen for particular mutations or conditions, e.g. carrier screening, prenatal screening, and newborn screening
- Stem cells – stem cells are harvested for their usefulness in being able to turn into other types of cells. Different types include adult, embryonic, or induced pluripotent stem cells
- Genetic modification – basically the same as gene therapy

Marking Criteria	*Mark(s)*
● Provides a judgement on the biotechnology, explaining how it works, its impact on society and advantages and disadvantages/limitations	5
● Explains how a biotechnology works, its impact on society and at least one advantage and one disadvantage/limitation	4
● Outlines the biotechnology and outlines advantages and disadvantages OR ● Explains how the biotechnology works and an advantage or disadvantage	3
● Outlines the biotechnology and one advantage or disadvantage	2
● Provides some relevant information	1